

 Marwadi University Marwadi Chandarana Group 	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	Aim: Generate Histogram of images, apply Histogram equalization and Histogram matching on it.	
OPE: 7	Date:24-08-2026	Enrollment No:92400133120

Aim: apply histogram

equalization and histogram matching techniques to enhance their visual quality.

Experiment:

We first load all the images using cv2 library and then convert them to gray scale image.

Then I have used two functions one for generating histplot for input image and another to show both input image along with it's plot. The function names are generate_histogram and plots respectively.

The code is below:

```
import cv2
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

input_img1=cv2.imread(r"C:\Users\bhoga\Downloads\Picture1.png")
input_img2=cv2.imread(r"C:\Users\bhoga\Downloads\Picture2.png")
input_img3=cv2.imread(r"C:\Users\bhoga\Downloads\Picture3.png")
input_img4=cv2.imread(r"C:\Users\bhoga\Downloads\Picture4.png")
input_img5=cv2.imread(r"C:\Users\bhoga\Downloads\Picture5.png")
gray_1=cv2.cvtColor(input_img1,cv2.COLOR_BGR2GRAY)
gray_2=cv2.cvtColor(input_img2,cv2.COLOR_BGR2GRAY)
gray_3=cv2.cvtColor(input_img3,cv2.COLOR_BGR2GRAY)
gray_4=cv2.cvtColor(input_img4,cv2.COLOR_BGR2GRAY)
gray_5=cv2.cvtColor(input_img5,cv2.COLOR_BGR2GRAY)
def generate_histogram(image):
    histogram=cv2.calcHist([image],[0],None,[256],[0,256])
    return histogram
hist1=generate_histogram(gray_1)
hist2=generate_histogram(gray_2)
hist3=generate_histogram(gray_3)
```

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```

hist4=generate_histogram(gray_4)
hist5=generate_histogram(gray_5)
def input_plots(gray,hist):
    intensity=np.arange(256)
    plt.figure(figsize=(6,4))
    plt.subplot(1,2,1)
    plt.imshow(gray,cmap='gray')
    plt.title("Input image")
    plt.axis('off')
    #hist plot for same image
    plt.subplot(1,2,2)
    plt.bar(intensity,hist)
    plt.title("Image histogram")
    plt.xlim([0,255])
    plt.xlabel("Intensity")
    plt.ylabel("Number of pixels")
    plt.tight_layout()
    return plt.show()

plot1=input_plots(gray_1,hist1)
plot2=input_plots(gray_2,hist2)
plot3=input_plots(gray_3,hist3)
plot4=input_plots(gray_4,hist4)
plot5=input_plots(gray_5,hist5)

```

Output of the code:

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Input image

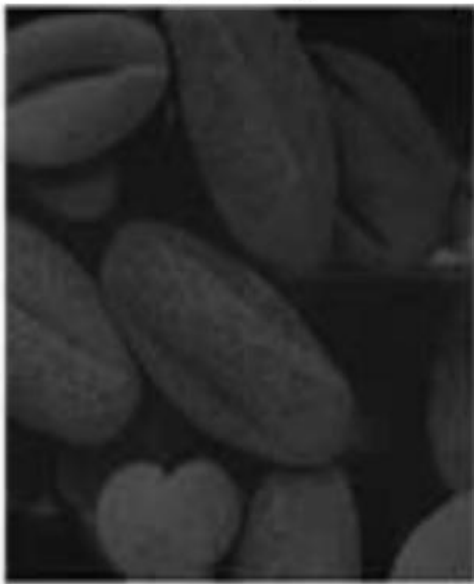
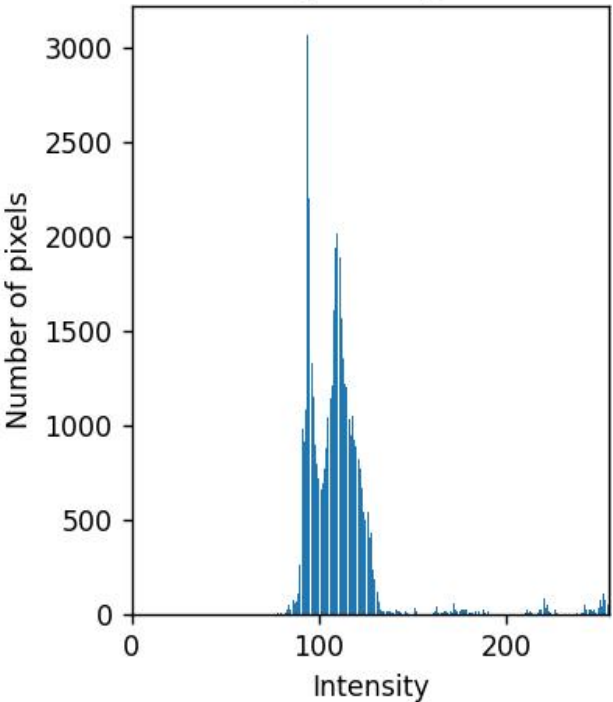


Image histogram



Input image

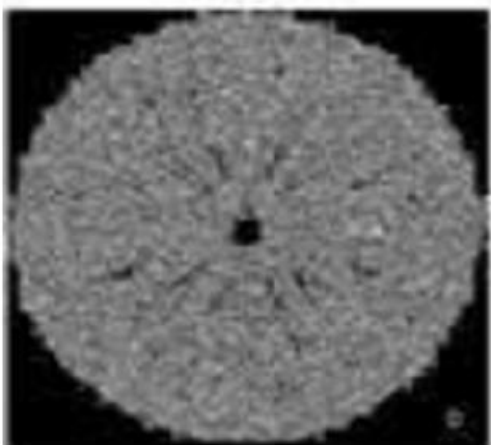
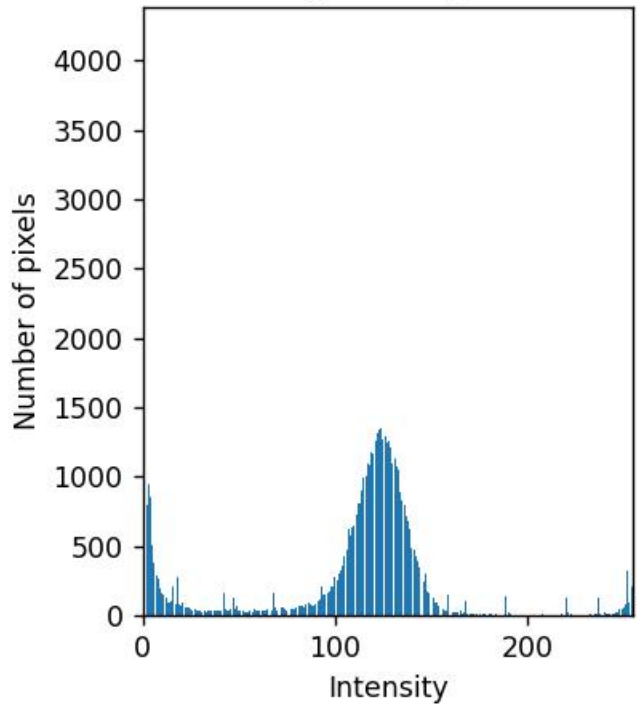
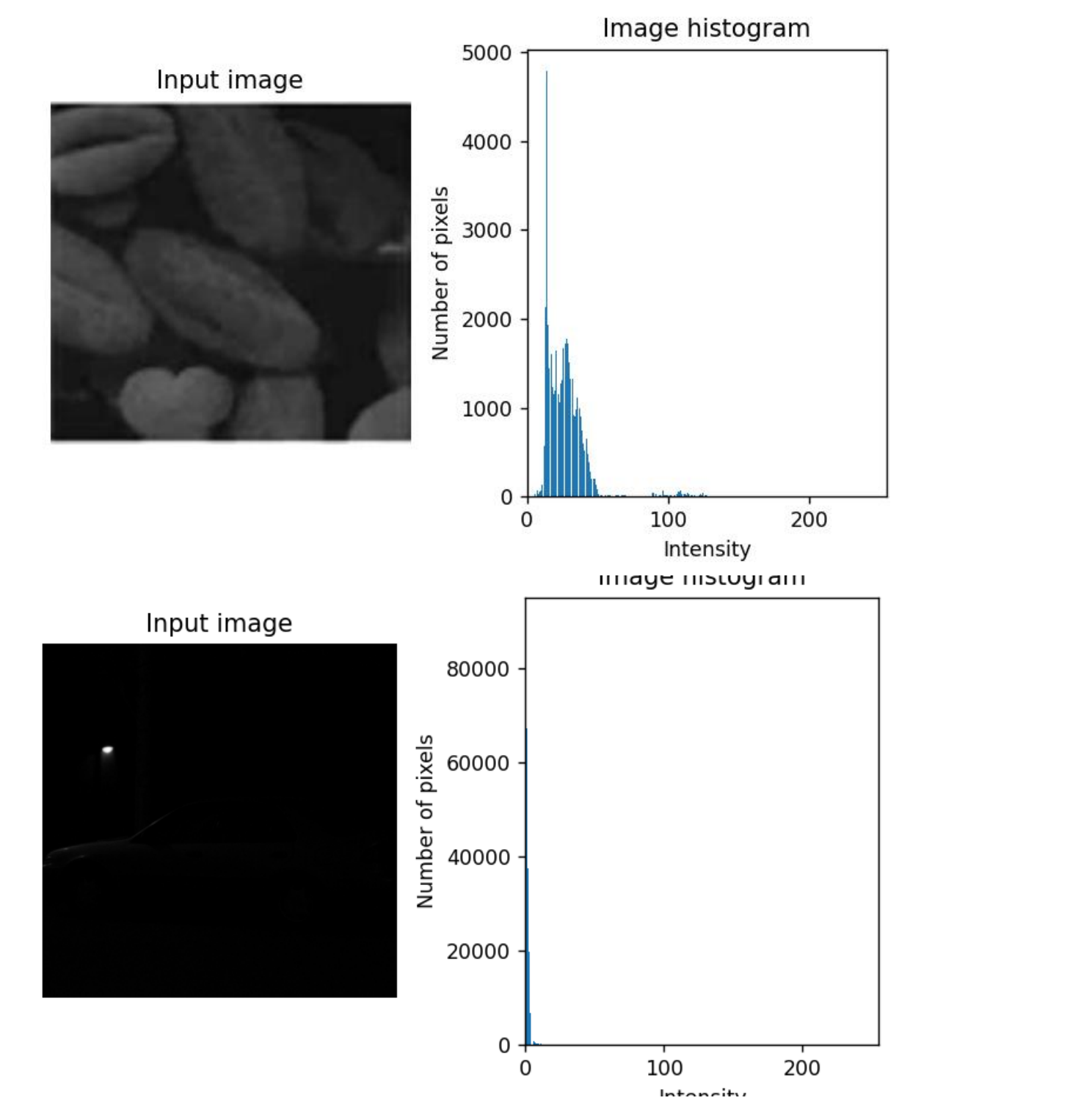


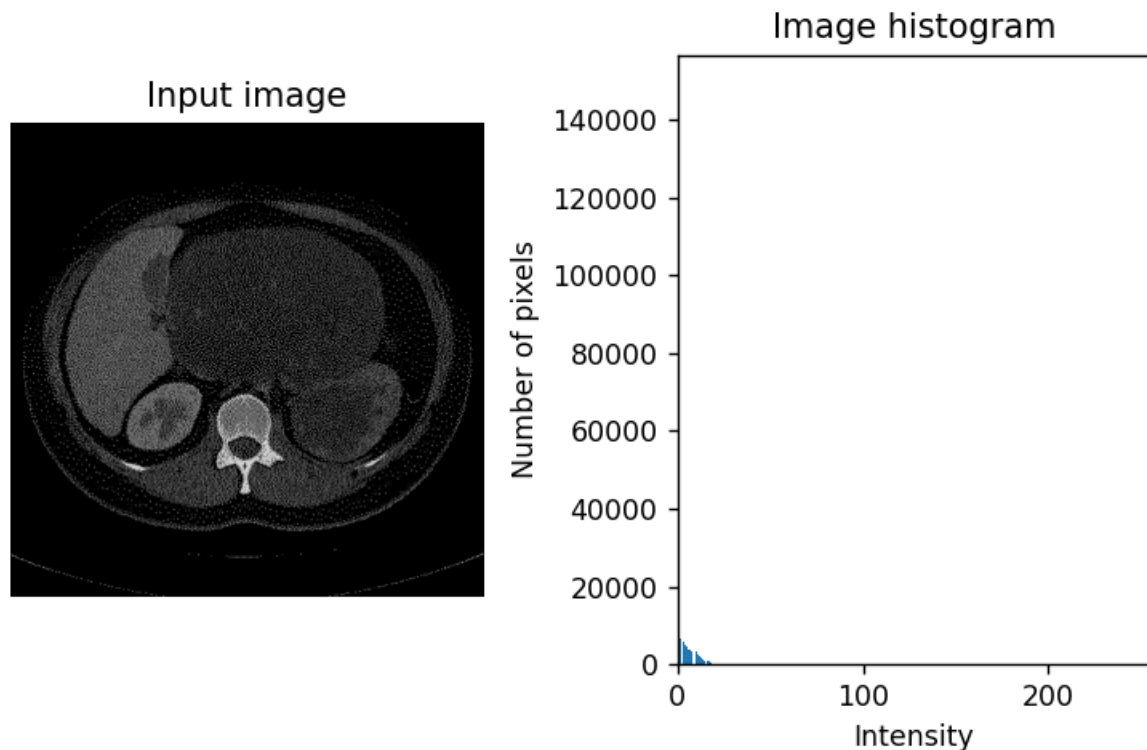
Image histogram



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Now we perform normalization on each image and see the effect of normalization on the image
Code:

```
def plot_eql_hist(image,hist,equal_image,eqlhist):
    intensity=np.arange(256)
    plt.figure(figsize=(8,6))
    plt.subplot(2,2,1)
    plt.imshow(image,cmap='gray')
    plt.title("Original image")
    plt.axis('off')
    plt.subplot(2,2,2)
    plt.bar(intensity,hist)
    plt.xlim([0,255])
    plt.xlabel("Intensity of pixels ")
    plt.ylabel('Number of pixels')
```

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```
plt.title("Original image histogram")
plt.subplot(2,2,3)
plt.imshow(eqval_image,cmap='gray')
plt.title("Equalized image")
plt.axis('off')
plt.subplot(2,2,4)
plt.bar(intensity,eqlhist)
plt.xlabel("equalized inimage")
plt.ylabel("Intensity")
plt.title("Histogram of equalized inimage")
plt.tight_layout()
return plt.show()

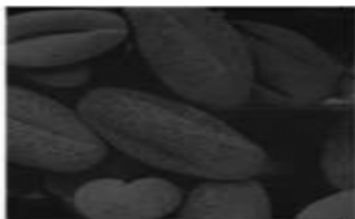
img1=plot_eql_hist(gray_1,hist1,eql_hist1,hist6)
img2=plot_eql_hist(gray_2,hist2,eql_hist2,hist7)
img3=plot_eql_hist(gray_3,hist3,eql_hist3,hist8)
img4=plot_eql_hist(gray_4,hist4,eql_hist4,hist9)
img5=plot_eql_hist(gray_5,hist5,eql_hist5,hist10)
```

Output:

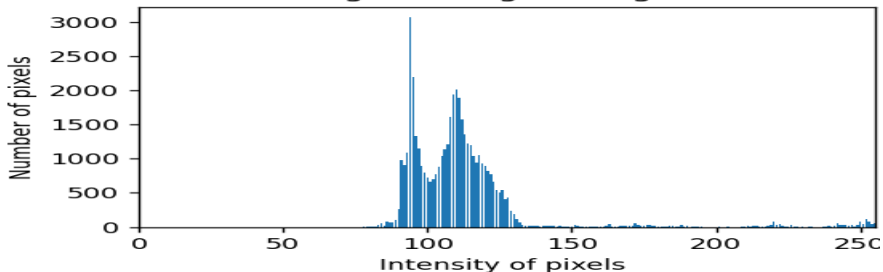
1st image:

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Original image



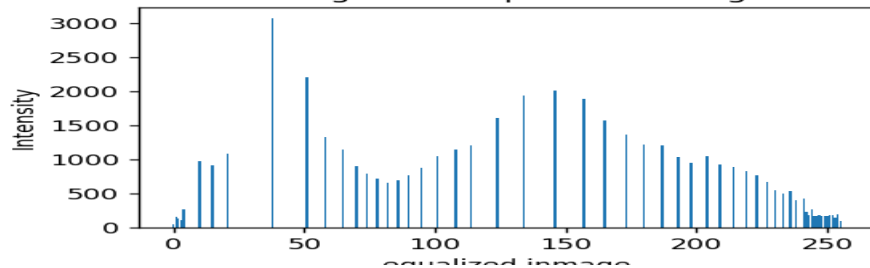
Original image histogram



Equalized image

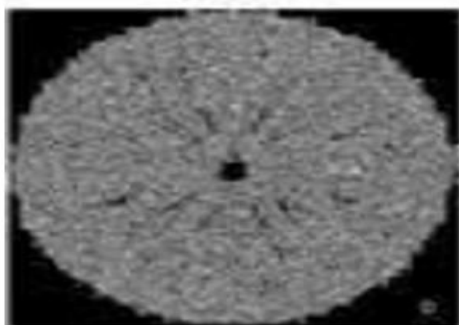


Histogram of equalized image

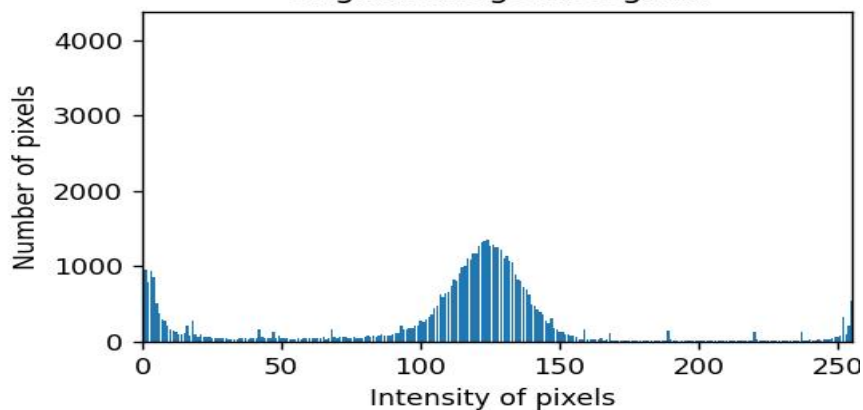


2nd image:

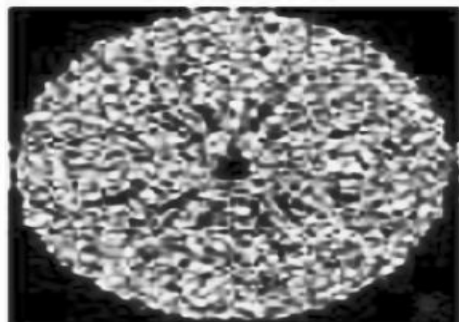
Original image



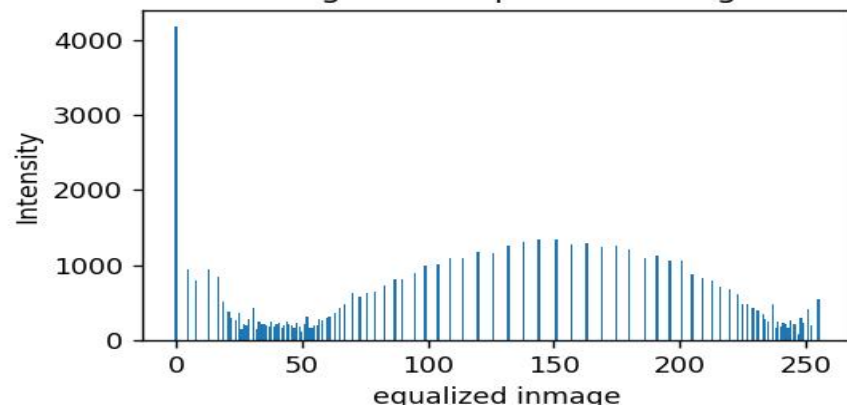
Original image histogram



Equalized image



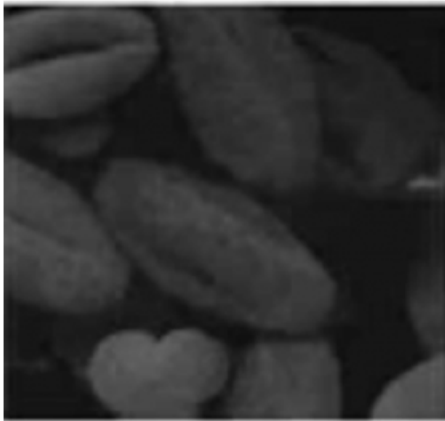
Histogram of equalized image



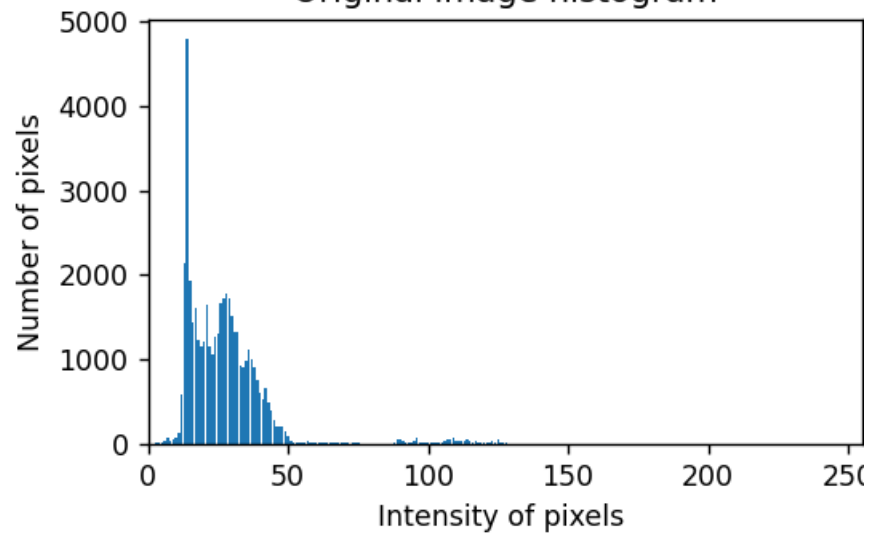
 Marwadi University Marwadi Chandarana Group	Marwadi University Faculty of Engineering & Technology Department of Information and Communication Technology	
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3rd image:

Original image



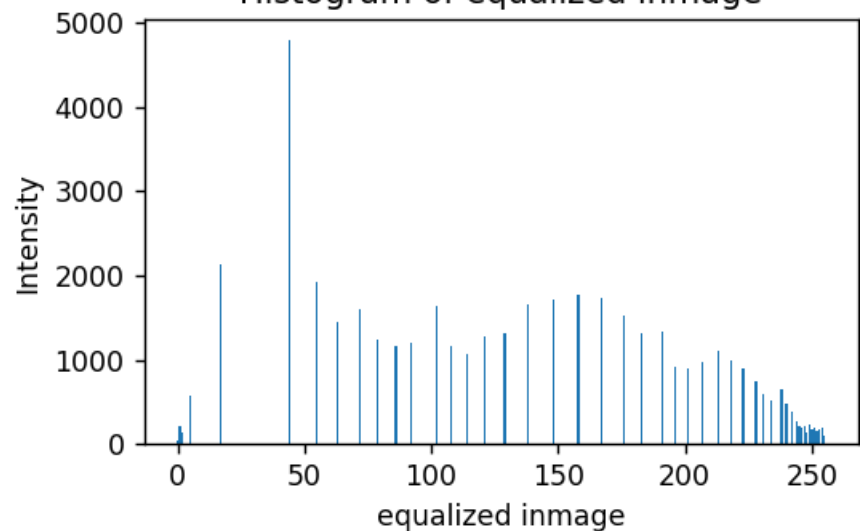
Original image histogram



Equalized image



Histogram of equalized image



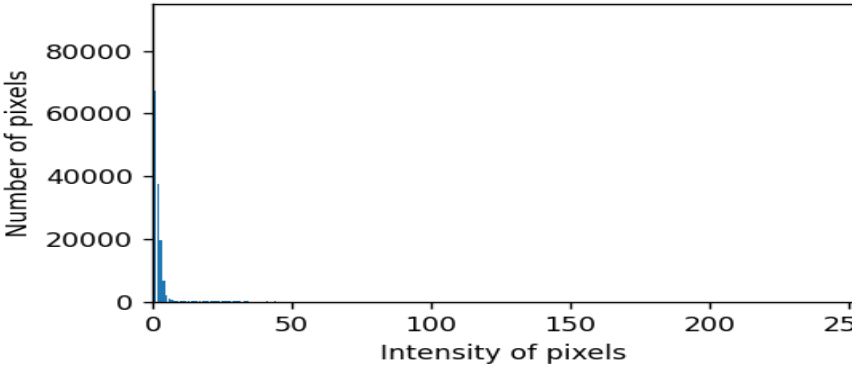
4th image:

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Original image



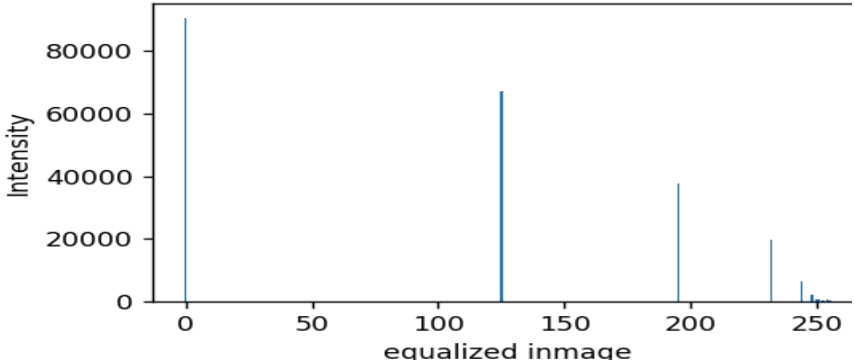
Original image histogram



Equalized image



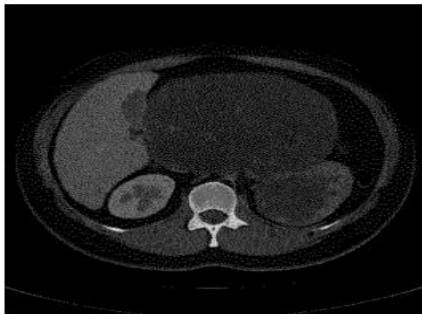
Histogram of equalized image



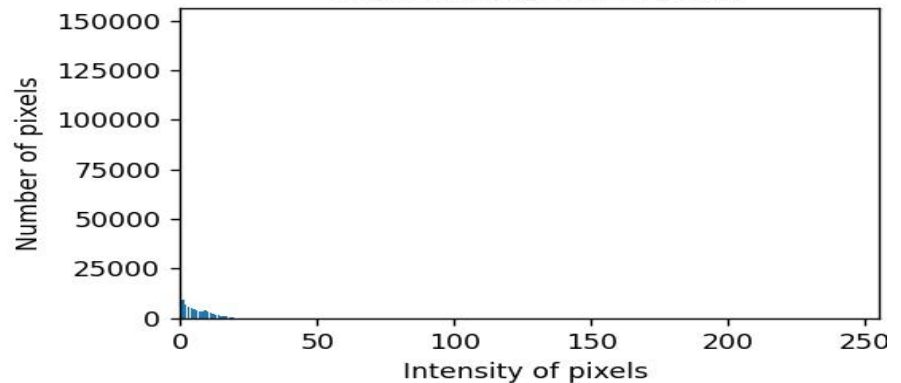
5th image:

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Original image



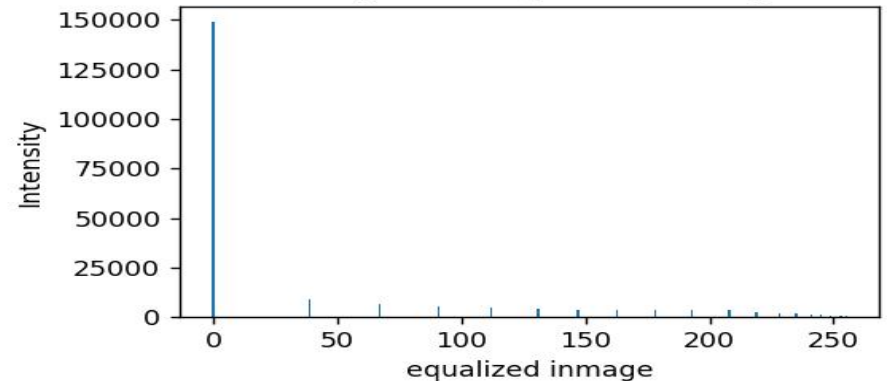
Original image histogram



Equalized image



Histogram of equalized inimage



Conclusion: As we can notice in the above images they became more enhanced and the details inside them became more clearer. As for 4th and 5th image a significant change can be observed. Here I have written a single function that plots the subgraph for equalized and original image.

Now we will map the original image with it's equalized image as it is normalized it can serve as how an ideal image should look. Here we will consider the first image as there the pixel intensity is distributed evenly and it can be observed by the histogram of the image.

Code:

```
from skimage.exposure import match_histograms
```

```
matched_img1=match_histograms(gray_1,eql_hist1)
```

```
matched_img2=match_histograms(gray_2,eql_hist1)
```

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```

matched_img3=match_histograms(gray_3,eql_hist1)
matched_img4=match_histograms(gray_4,eql_hist1)
matched_img5=match_histograms(gray_5,eql_hist1)
m_img1=np.uint8(matched_img1)
m_img2=np.uint8(matched_img2)
m_img3=np.uint8(matched_img3)
m_img4=np.uint8(matched_img4)
m_img5=np.uint8(matched_img5)
m_hist1=generate_histogram(m_img1)
m_hist2=generate_histogram(m_img2)
m_hist3=generate_histogram(m_img3)
m_hist4=generate_histogram(m_img4)
m_hist5=generate_histogram(m_img5)

```

```

def plot_match_histogram(image,r_image,hist,r_hist,m_image,m_hist):
    intensity=np.arange(256)
    plt.figure(figsize=(8,6))
    plt.subplot(3,2,1)
    plt.imshow(image,cmap='gray')
    plt.title("Original image")
    plt.axis('off')
    plt.subplot(3,2,2)
    plt.bar(intensity,hist)
    plt.xlabel("Pixel range")
    plt.ylabel("Frequency")
    plt.title("Hist plaot of original image")
    plt.xlim([0,256])
    plt.subplot(3,2,3)

```

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```

plt.imshow(r_image,cmap='gray')
plt.title("reference image")
plt.axis('off')
plt.subplot(3,2,4)
plt.bar(intensity,r_hist)
plt.xlabel("Pixel range")
plt.ylabel("Frequency")
plt.title("Hist plaot of reference image")
plt.xlim([0,256])
plt.subplot(3,2,5)
plt.imshow(m_image,cmap='gray')
plt.title("matched image")
plt.axis('off')
plt.subplot(3,2,6)
plt.bar(intensity,m_hist)
plt.xlabel("Pixel range")
plt.ylabel("Frequency")
plt.title("Hist plaot of matched image")
plt.xlim([0,256])
plt.tight_layout()
return plt.show()

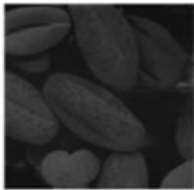
plt1=plot_match_histogram(gray_1,eql_hist1,hist1,hist6,m_img1,m_hist1)
plt2=plot_match_histogram(gray_2,eql_hist1,hist2,hist6,m_img2,m_hist2)
plt3=plot_match_histogram(gray_3,eql_hist1,hist3,hist6,m_img3,m_hist3)
plt4=plot_match_histogram(gray_4,eql_hist1,hist4,hist6,m_img4,m_hist4)
plt5=plot_match_histogram(gray_5,eql_hist1,hist5,hist6,m_img5,m_hist5)

```

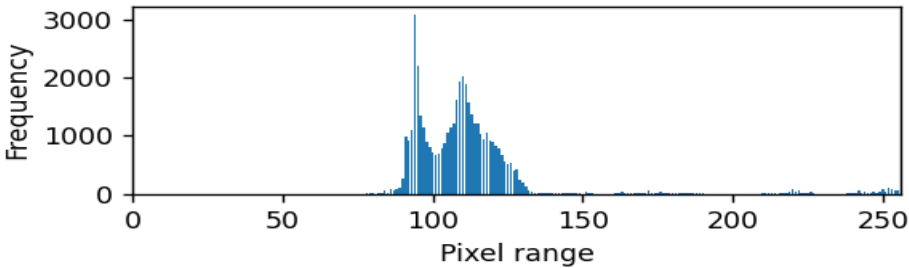
Output:

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Original image



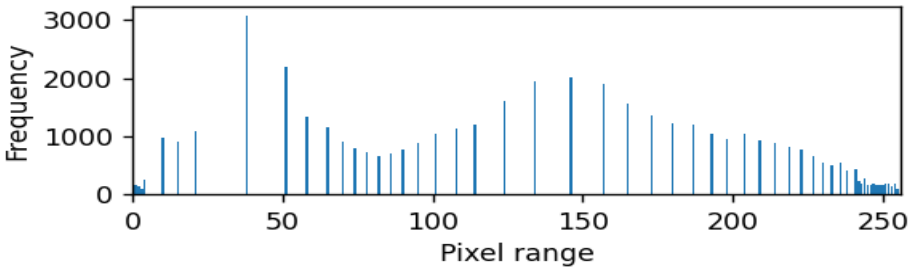
Hist ploat of original image



reference image



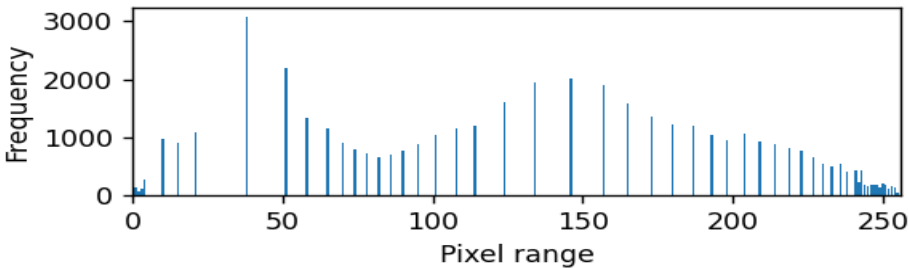
Hist ploat of reference image



matched image

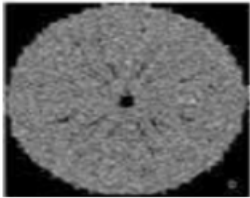


Hist ploat of matched image

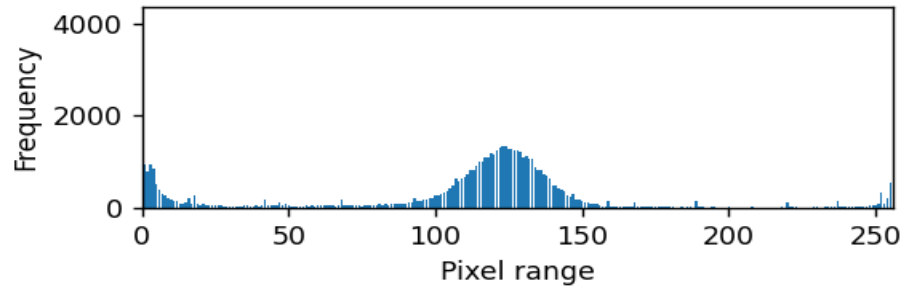


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Original image



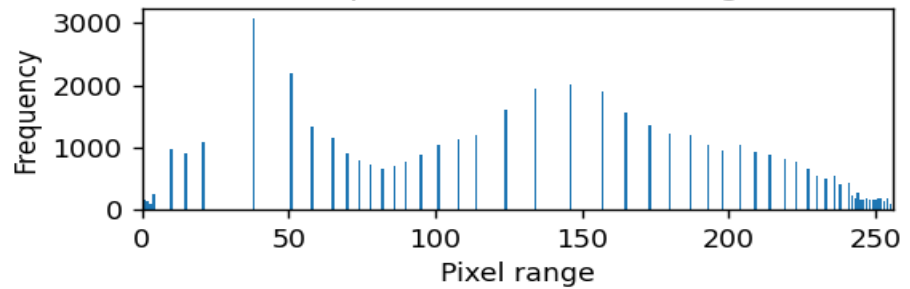
Hist plaot of original image



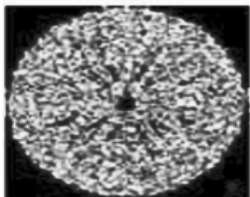
reference image



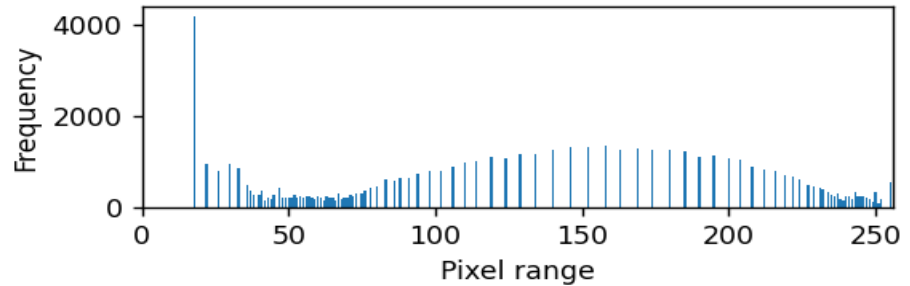
Hist plaot of reference image



matched image

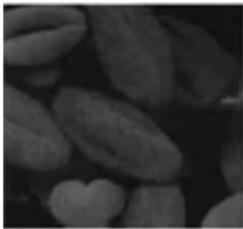


Hist plaot of matched image

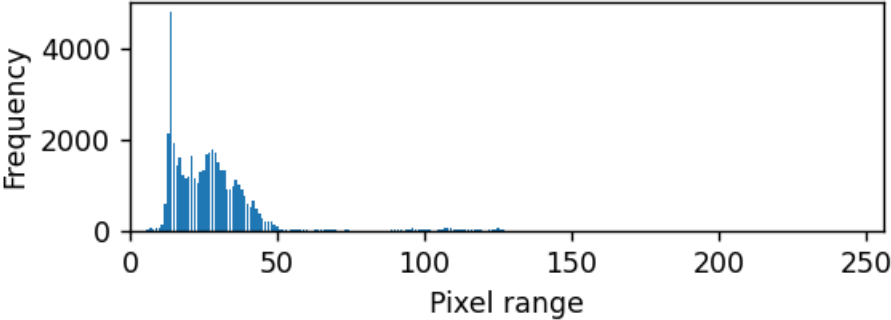


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Original image



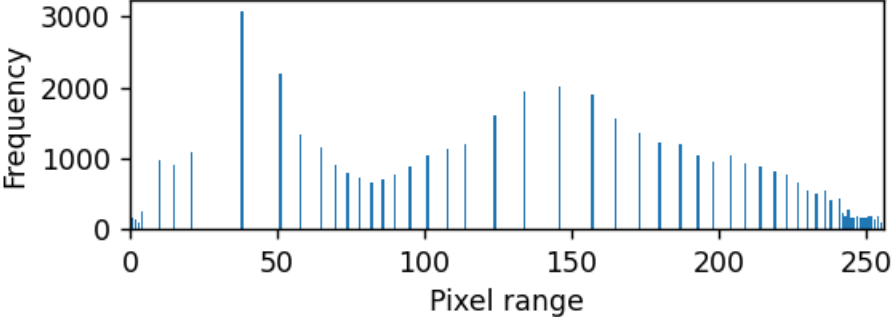
Hist ploat of original image



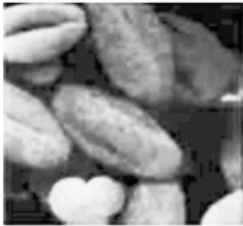
reference image



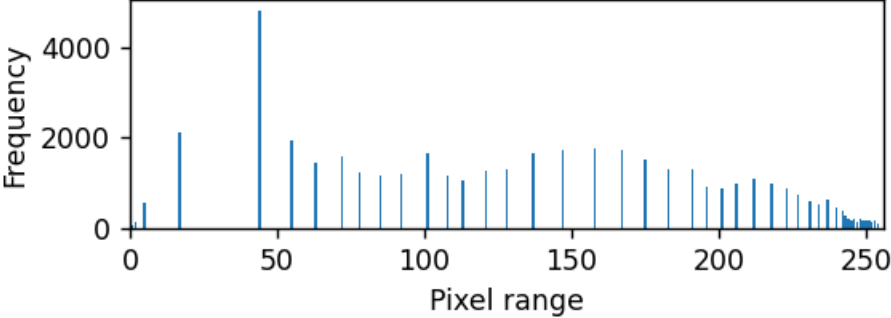
Hist ploat of reference image



matched image

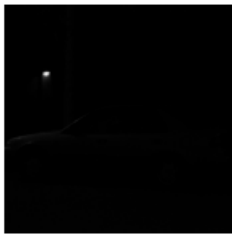


Hist ploat of matched image

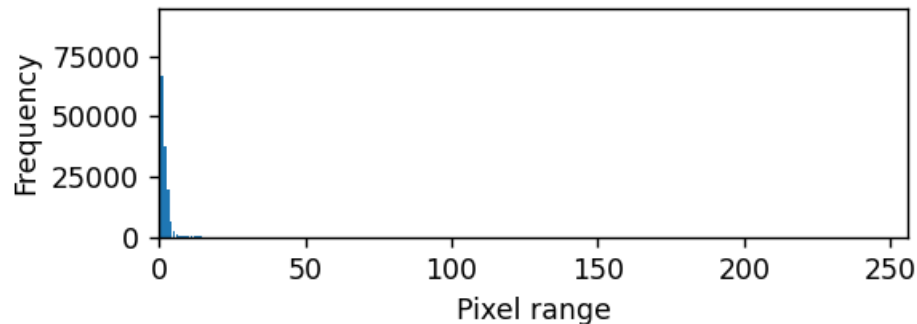


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Original image



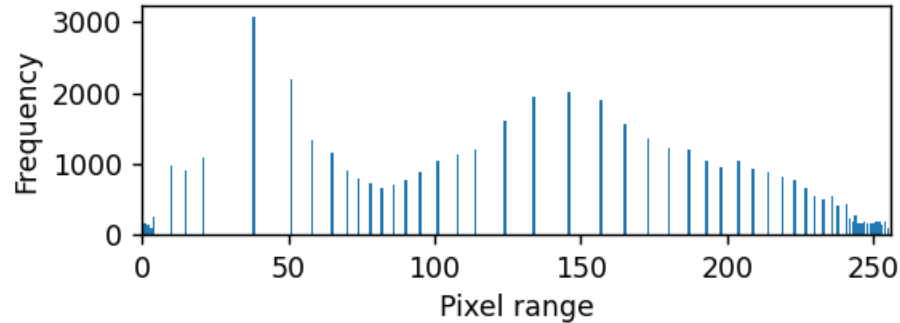
Hist plaot of original image



reference image



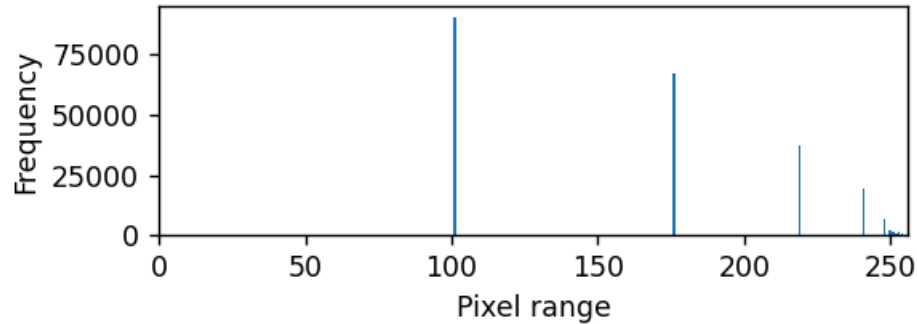
Hist plaot of reference image



matched image

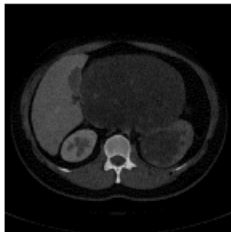


Hist plaot of matched image

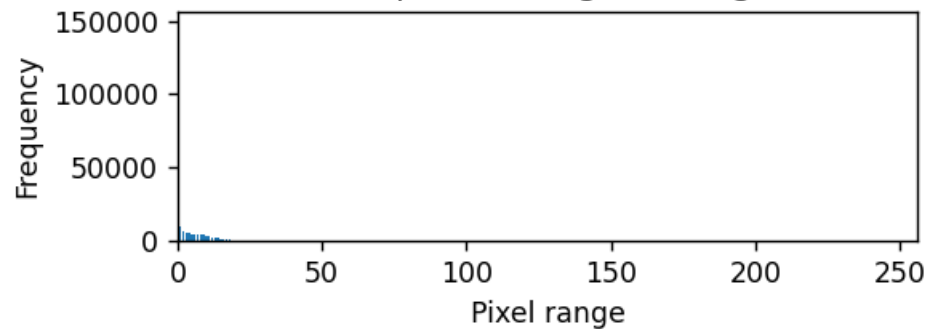


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Original image



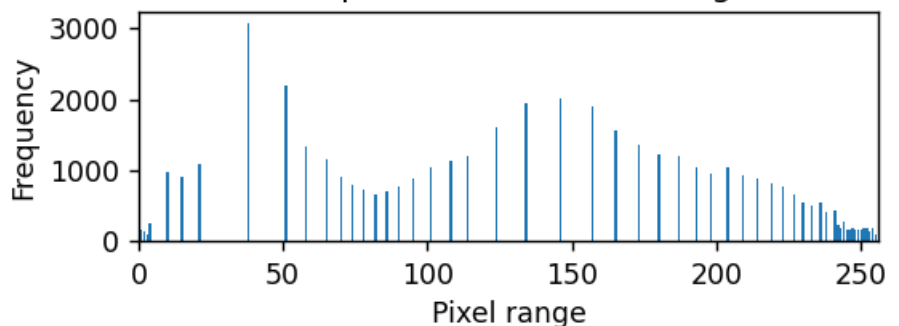
Hist ploat of original image



reference image



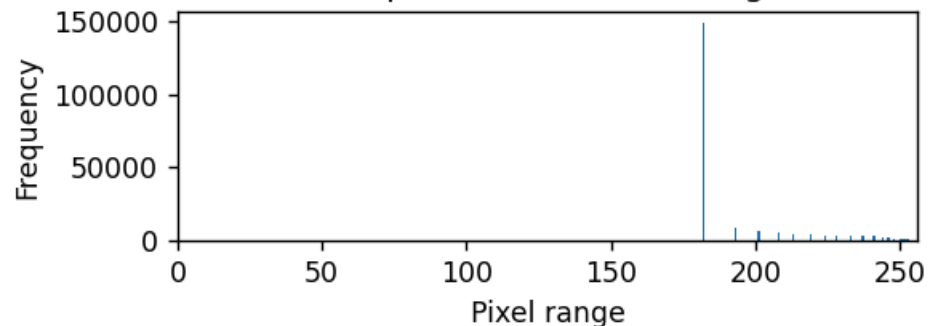
Hist ploat of reference image



matched image



Hist ploat of matched image



Conclusion:

Here we can observe that we get a more clear image when we match two images. However the main condition to be satisfied here is that your reference image must have evenly spreaded out pixels or it should have good context. Here batter results were obtained as I took an already equalised image. If the reference image it self has uneven distribution of pixels then the resulting image may not have desired output. Hence hist equalization is a batter option even thought we are getting more fine results in matching method.

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